

Fundamentals of AI and KR - Module 3

3. Building Bayesian networks

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Notice

Credits

The present slides are largely an adaptation of existing material, including:

- slides from [Russel & Norvig](#)
- slides by [Daphne Koller](#) on Probabilistic Graphical Models
- slides by Fabrizio Riguzzi on [Data Mining and Analytics](#)

I am especially grateful to these authors.

Downloading and sharing

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- Causal networks
- Representing conditional distributions

Constructing Bayesian networks



Constructing Bayesian networks

Need a method such that a series of locally testable assertions of conditional independence guarantees the required global semantics

- ① Choose an ordering of variables $X_1, \dots, X_n$
- ② For $i = 1$ to n
 - add X_i to the network
 - select parents from $X_1, \dots, X_{i-1}$ such that

$$\mathbf{P}(X_i | \text{Parents}(X_i)) = \mathbf{P}(X_i | X_1, \dots, X_{i-1})$$

This choice of parents guarantees the global semantics:

$$\begin{aligned}
 \mathbf{P}(X_1, \dots, X_n) &= \prod_{i=1}^n \mathbf{P}(X_i | X_1, \dots, X_{i-1}) \quad (\text{chain rule}) \\
 &= \prod_{i=1}^n \mathbf{P}(X_i | \text{Parents}(X_i)) \quad (\text{by construction})
 \end{aligned}$$



Example

Let's construct a Bayesian network that helps us win a prize

Monty Hall puzzle

We're **guests** on a TV game show. We stand in front of three closed doors. A **prize** hides behind one of them. We choose the door on the left. At this point, the **host**, who knows where prize is, opens the middle door, to reveal it is empty. We are offered to modify our choice. *Should we?*



Example

Suppose we choose the ordering M, J, A, B, E

MaryCalls

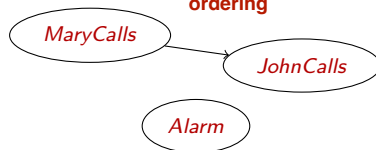
JohnCalls

- $P(J|M) = P(J)$?



Example

Suppose we choose the ordering M, J, A, B, E
 ordering $\rightarrow$



Network strongly
depends on ordering



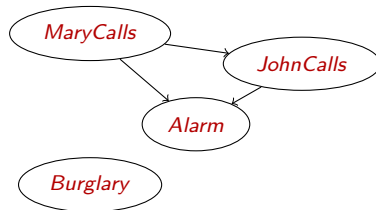
- $P(J|M) = P(J)$? No
- $P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$?

J is not independent from M and from ordering we assume
M directly influences J



Example

Suppose we choose the ordering M, J, A, B, E

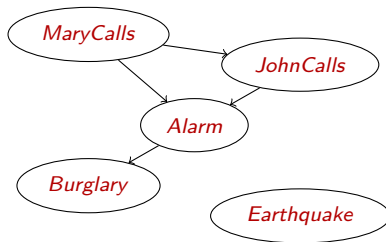


- $P(J|M) = P(J)$? No
- $P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$? No
- $P(B|A, J, M) = P(B|A)$?
- $P(B|A, J, M) = P(B)$?



Example

Suppose we choose the ordering M, J, A, B, E

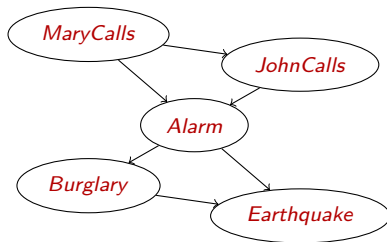


- $P(J|M) = P(J)$? No
- $P(A|J, M) = P(A|J)$? $P(A|J, M) = P(A)$? No
- $P(B|A, J, M) = P(B|A)$? Yes
- $P(B|A, J, M) = P(B)$? No
- $P(E|B, A, J, M) = P(E|A)$?
- $P(E|B, A, J, M) = P(E|A, B)$?



Example

Suppose we choose the ordering M, J, A, B, E

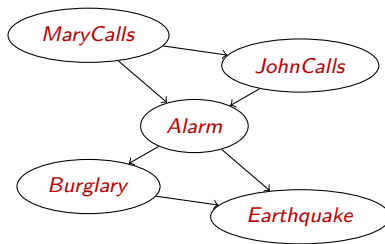


- $P(J|M) = P(J)$? No
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- $P(E|B, A, J, M) = P(E|A)$? No
- $P(E|B, A, J, M) = P(E|A, B)$? Yes



Example

Suppose we choose the ordering M, J, A, B, E

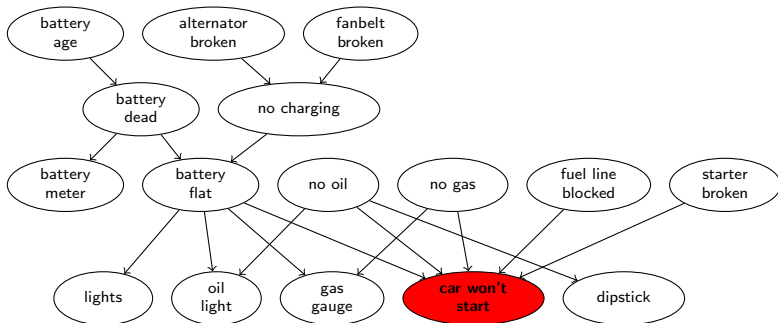


Deciding conditional independence is **hard in noncausal directions**
(Causal models and conditional independence seem hardwired for humans!)
Assessing conditional probabilities is hard in noncausal directions
Network is less compact: $1 + 2 + 4 + 2 + 4 = 13$ numbers needed



Example: Car diagnosis

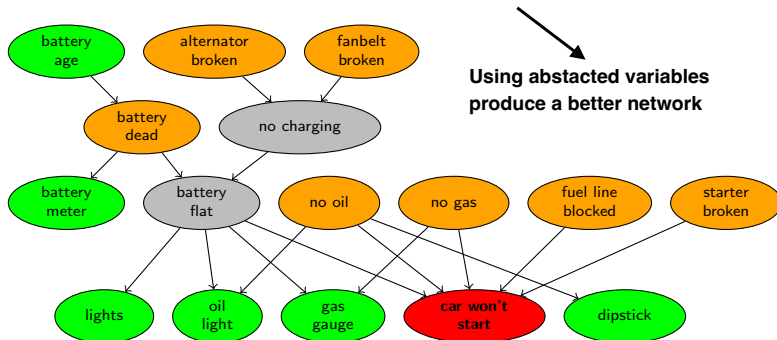
- Initial evidence: car won't start





Example: Car diagnosis

- Initial evidence: car won't start
- Testable variables (green), "broken, so fix it" variables (orange)
- Hidden variables (gray) ensure sparse structure, reduce parameters



Structure learning

If manual design not possible, learn the network from the available data.

Two approaches:

- Constraint-based **Network given + data = full network**
 - ① **independence test** to identify a set of edge constraints for the graph
 - ② search to best satisfy the constraints
- Score-based: **data + data --> network = full network** **Trying different orders of variables**
 - ① **define a criterion** (score) to evaluate how well the Bayesian network fits the data,
 - ② search to maximize score
- Obtaining good quality results may be tricky
- Use domain knowledge if available

Causal networks



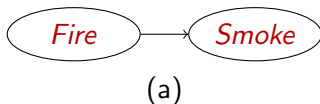
Causal networks

- In principle, any ordering of nodes permits a consistent construction of the network
- However, best to respect order of causality for many reasons, including compactness
- **Causal networks** are a restricted class of Bayesian networks that forbids all but causally compatible orderings



Causal networks

- Consider the following example:

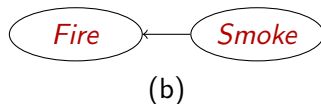
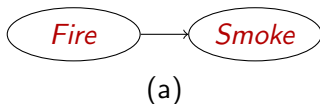




Causal networks

- Consider the following example:

In order to identify which is causal we can answer to “which responds to which?”, the idea is that Fire causes smoke, so smoke responds to fire and so (a) is causal.



- Equally good distributions can be defined for (a) and (b)
 - But *are these networks equivalent?*



Beyond probabilistic dependence: assignment

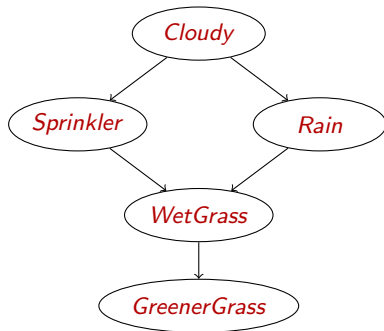
- Causal networks devised to represent causal asymmetries
- Arrow directionality decided based on considerations beyond probabilistic dependence
- The question is: **which responds to which?**
 - Draw an arrow from **Fire** to **Smoke** if nature “assigns” a value to **Smoke** on the basis of what nature learns about **Fire**
 - Do not draw an arrow from **Smoke** to **Fire** if you judge that nature “assigns” a **Fire** a truth value based on variables other than **Smoke**
 - For each variable X_i that can take values $x_i = f_i(\text{OtherVariables})$, draw $X_j \rightarrow X_i$ if and only if X_j is one of the arguments of f_i

$x_i = f_i(\cdot)$ is called a **structural equation**



Lawn example

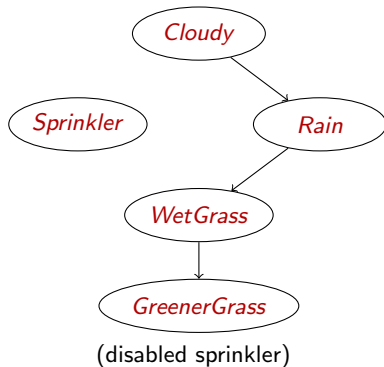
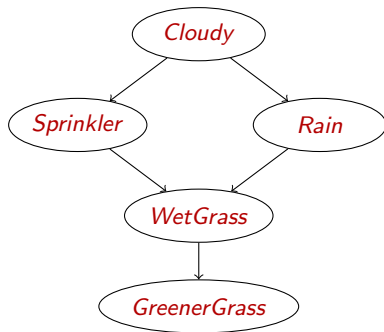
- Structural equations describe mechanism in nature invariant to measurements and local changes in the environment





Lawn example

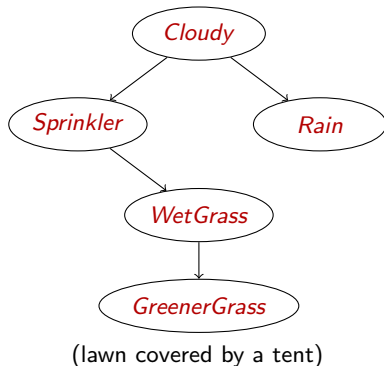
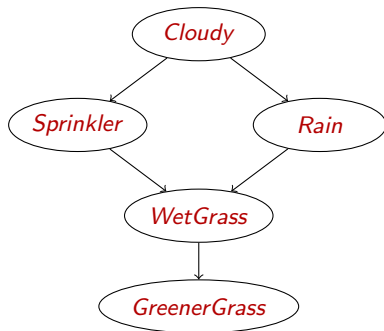
- Structural equations describe mechanism in nature invariant to measurements and local changes in the environment





Lawn example

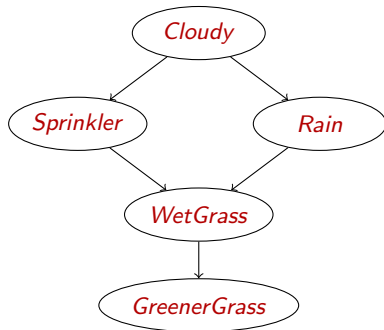
- Structural equations describe mechanism in nature invariant to measurements and local changes in the environment





The *do*-operator

- Stability is important for representing **interventions** and predicting their observable consequences



- Semantics of Bayes nets:

$$P(c, r, s, w, g)$$

- System of structural equations with “*U*-variables” (**unmodeled variables** or **error terms**):

$$C = f_C(U_C)$$

$$R = f_R(C, U_R)$$

$$S = f_S(C, U_S)$$

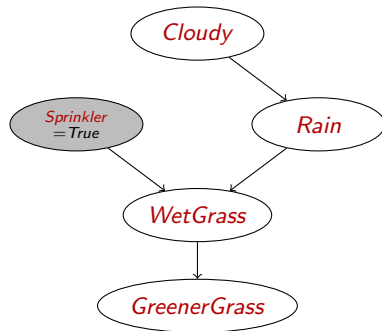
$$W = f_W(S, R, U_W)$$

$$G = f_G(W, U_G)$$



The *do*-operator

- Stability is important for representing **interventions** and predicting their observable consequences



$do(\text{Sprinkler} = \text{True})$:
we turn the sprinkler on

- Semantics of Bayes nets:

$$P(c, r, w, g | do(S = \text{true}))$$

- System of structural equations with “*U*-variables” (**unmodeled variables** or **error terms**):

$$C = f_C(U_C)$$

$$R = f_R(C, U_R)$$

$$S = \text{True}$$

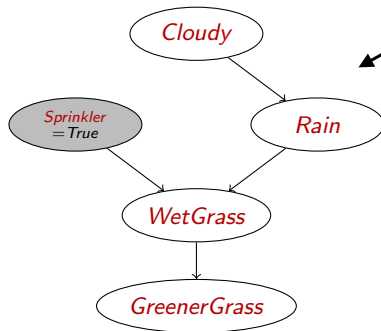
$$W = f_W(S, R, U_W)$$

$$G = f_G(W, U_G)$$

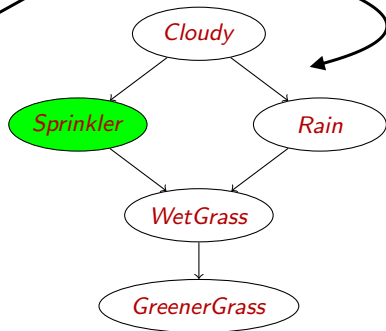


The *do*-operator

- What difference? ^(conceptual) Difference between making hypothesis and observing



$P(\text{WetGrass} | \text{do}(\text{Sprinkler} = \text{True}))$:
we **turn** the sprinkler on



$P(\text{WetGrass} | \text{Sprinkler} = \text{True})$:
we **observe** that sprinkler is on

Representing conditional distributions



Compact conditional distributions

CPT grows exponentially with number of parents

CPT becomes infinite with continuous-valued parent or child

Solution: **canonical** distributions that are defined compactly

Deterministic nodes are the simplest case:

$X = f(\text{Parents}(X))$ for some function f

e.g., Boolean functions

$\text{NorthAmerican} \Leftrightarrow \text{Canadian} \vee \text{US} \vee \text{Mexican}$

e.g., numerical relationships among continuous variables

$$\frac{\partial \text{Level}}{\partial t} = \text{inflow} + \text{precipitation} - \text{outflow} - \text{evaporation}$$

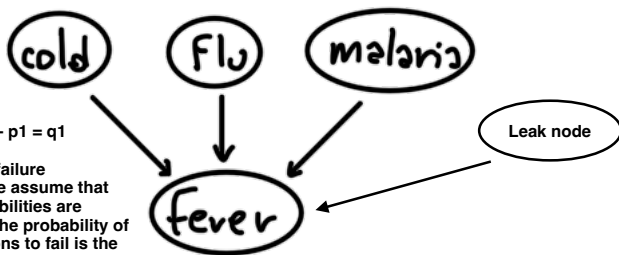


Compact conditional distributions

Noisy-OR distributions model multiple noninteracting causes

- ① Parents $U_1 \dots U_k$ include all causes (can add leak node)
- ② Independent failure probability q_i for each cause alone

$$\Rightarrow P(X|U_1 \dots U_j, \neg U_{j+1} \dots \neg U_k) = 1 - \prod_{i=1}^j q_i$$



$P(\text{fever} | \text{cold}) = p_1$
 $P(\neg \text{fever} | \text{cold}) = 1 - p_1 = q_1$

The last one is the failure probability and if we assume that all the failure probabilities are independent, then the probability of failing of N conditions to fail is the product of all the failure probability.



Compact conditional distributions

Noisy-OR distributions model multiple noninteracting causes

① Parents $U_1 \dots U_k$ include all causes (can add leak node)

② Independent failure probability q_i for each cause alone

$$\Rightarrow P(X|U_1 \dots U_j, \neg U_{j+1} \dots \neg U_k) = 1 - \prod_{i=1}^j q_i$$

<i>Cold</i>	<i>Flu</i>	<i>Malaria</i>	$P(\text{Fever})$	$P(\neg \text{Fever})$
F	F	F	0.0	1-0.0 = 1
F	F	T	1-0.1=0.9	0.1
F	T	F	1-0.2=0.8	0.2
F	T	T	1-0.6=0.4	0.6
T	F	F		
T	F	T		
T	T	F		
T	T	T		



Compact conditional distributions

Noisy-OR distributions model multiple noninteracting causes

① Parents $U_1 \dots U_k$ include all causes (can add **leak node**)

② Independent failure probability q_i for each cause alone

$$\Rightarrow P(X|U_1 \dots U_j, \neg U_{j+1} \dots \neg U_k) = 1 - \prod_{i=1}^j q_i$$

<i>Cold</i>	<i>Flu</i>	<i>Malaria</i>	$P(\text{Fever})$	$P(\neg \text{Fever})$
F	F	F	0.0	0.1 0.2 0.6
F	F	T		
F	T	F		
F	T	T		
T	F	F		
T	F	T		
T	T	F		
T	T	T		

Number of parameters linear in number of parents



Compact conditional distributions



Noisy-OR distributions model multiple noninteracting causes

① Parents $U_1 \dots U_k$ include all causes (can add leak node)

② Independent failure probability q_i for each cause alone

$$\Rightarrow P(X|U_1 \dots U_j, \neg U_{j+1} \dots \neg U_k) = 1 - \prod_{i=1}^j q_i$$

	<i>Cold</i>	<i>Flu</i>	<i>Malaria</i>	$P(\text{Fever})$	$P(\neg \text{Fever})$
P(Malaria)	F	F	F	0.0	
	F	F	T		0.1
	F	T	F		0.2
P(Flu)	F	T	T		$0.02 = 0.2 \times 0.1$
P(Cold)	T	F	F		0.6
	T	F	T		$0.06 = 0.6 \times 0.1$
	T	T	F		$0.12 = 0.6 \times 0.2$
	T	T	T		$0.012 = 0.6 \times 0.2 \times 0.1$



Compact conditional distributions

Noisy-OR distributions model multiple noninteracting causes

① Parents $U_1 \dots U_k$ include all causes (can add leak node)

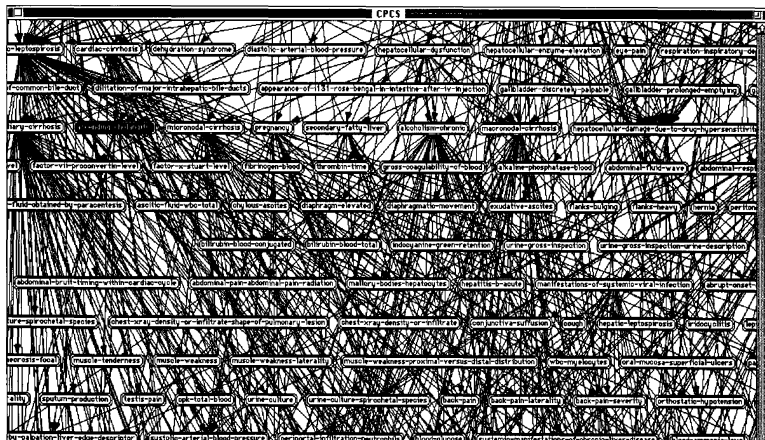
② Independent failure probability q_i for each cause alone

$$\Rightarrow P(X|U_1 \dots U_j, \neg U_{j+1} \dots \neg U_k) = 1 - \prod_{i=1}^j q_i$$

<i>Cold</i>	<i>Flu</i>	<i>Malaria</i>	$P(\text{Fever})$	$P(\neg \text{Fever})$
F	F	F	0.0	1.0
F	F	T	0.9	0.1
F	T	F	0.8	0.2
F	T	T	0.98	$0.02 = 0.2 \times 0.1$
T	F	F	0.4	0.6
T	F	T	0.94	$0.06 = 0.6 \times 0.1$
T	T	F	0.88	$0.12 = 0.6 \times 0.2$
T	T	T	0.988	$0.012 = 0.6 \times 0.2 \times 0.1$

A Bayesian network for internal medicine

Knowledge Engineering for Large Belief Networks, Pradhan et al., UAI 1994



A Bayesian network for internal medicine

Knowledge Engineering for Large Belief Networks, Pradhan et al., UAI 1994

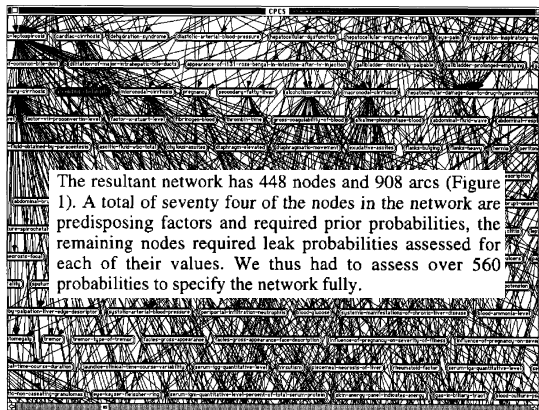
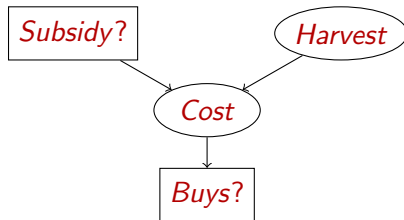


Figure 1. A small portion of the CPCS BN displayed in the Netview visualization program. The node *ascending-cholangitis* in the third row shown in inverse has been selected by the user.



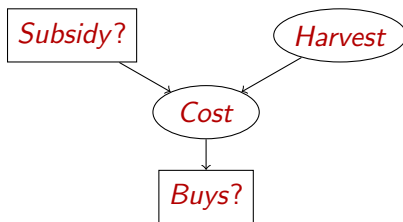
Hybrid (discrete+continuous) networks





Hybrid (discrete+continuous) networks

Discrete (*Subsidy?* and *Buys?*), continuous (*Harvest* and *Cost*)



- Option 1: discretization—possibly large errors, large CPTs To do in pgmpy
- Option 2: finitely parameterized canonical families
 - Continuous variable, discrete+continuous parents (e.g., *Cost*)
 - Discrete variable, continuous parents (e.g., *Buys?*)



Continuous child variables

Need one **conditional density** function for child variable given continuous parents, for each possible assignment to discrete parents

Most common is the **linear Gaussian** model, e.g.,:

$$\begin{aligned} P(\text{Cost} = c | \text{Harvest} = h, \text{Subsidy?} = \text{true}) \\ = \mathcal{N}(a_t h + b_t, \sigma_t)(c) = \frac{1}{\sigma_t \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{c - (a_t h + b_t)}{\sigma_t} \right)^2} \end{aligned}$$

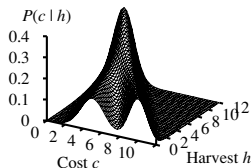
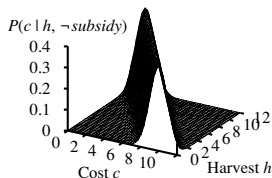
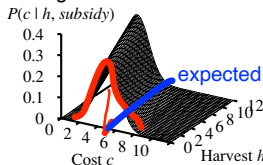
Mean **Cost** varies linearly with **Harvest**, variance is fixed

Linear variation is unreasonable over the full range but works OK if the likely range of **Harvest** is narrow



Continuous child variables

The mean of the gaussian changes wrt the Harvest



All-continuous network with LG distributions

⇒ full joint distribution is a multivariate Gaussian

Discrete parents added to continuous variables

⇒ LG network is a **conditional Gaussian** network

i.e., a multivariate Gaussian over all continuous variables for each combination of discrete variable values

e.g., $\mathbf{P}(\text{Cost} | \text{Harvest})$ obtained by summing over subsidy cases



Discrete variable w / continuous parents

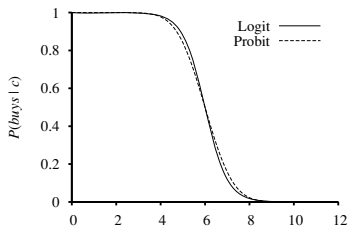
Probability of *Buys?* given *Cost* should be a “soft” threshold

Probit or sigmoid (logit) distribution

e.g., sigmoid: $P(\text{Buys?} = \text{true} \mid \text{Cost} = c) = \frac{1}{1 + \exp(-2 \frac{-c + \mu}{\sigma})}$

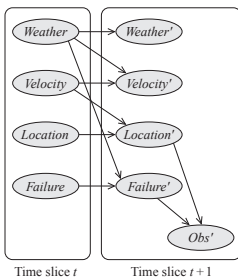
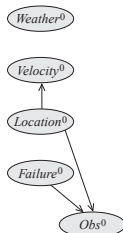
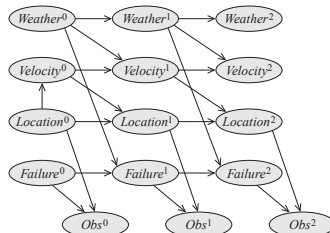
Probit uses integral of Gaussian

Sigmoid has similar shape to probit
but much longer tails



Template-based representations

Especially useful for reasoning about world that evolves over time
System state at time t , $\mathcal{X}^{(t)}$ $X_i \in \mathcal{X}$ is a **template variable**, instantiated at each time. $X_i^{(t)}$ is a variable. Each “possible world” is a **trajectory**
 Goal: to represent a joint distribution over trajectories

(a) $\mathcal{B}_{\rightarrow}$ (b) $\mathcal{B}_0$ 

(c) DBN unrolled over 3 steps

Density estimation

Parameters (conditional distributions) can be elicited by domain experts, imposed by the environment or model, or learned from experience

- Many methods for learning distributions from data (**density estimation**). Idea is:
 - Data are **evidence**,
 - **Hypotheses** are probabilistic theories about the domain
- **Bayesian learning** calculates the probability of each hypothesis, given the data
 - Predictions made using *all* hypotheses, weighted by their probabilities
 - Learning is probabilistic inference
 - Optimal prediction, but computationally expensive process
- Approximations: **MAP hypothesis** (keeps only one theory) and **maximum-likelihood hypothesis** (assumes uniform prior)
- **EM** algorithm for learning with hidden variables

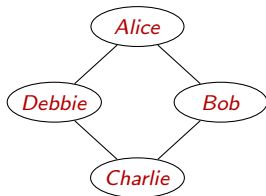
See Russel & Norvig, 4th Ed, Chapter 21 *Learning Probabilistic Models*

Undirected Graphical Models

Bayesian networks are a type of **Probabilistic Graphical Models** (PGM)

Another class of PGMs are **Markov networks**

- undirected
- factors, not CPT (do not represent probability distributions) levels of AFFINITY
- naturally capture conditional independence relations



Misconception network

$\phi_1(A, B)$			$\phi_2(B, C)$			$\phi_3(C, D)$			$\phi_4(D, A)$		
a^0	b^0	30	b^0	c^0	100	c^0	d^0	1	d^0	a^0	100
a^0	b^1	5	b^0	c^1	1	c^0	d^1	100	d^0	a^1	1
a^1	b^0	1	b^1	c^0	1	c^1	d^0	100	d^1	a^0	1
a^1	b^1	10	<u>b^1</u>	<u>c^1</u>	<u>100</u>	c^1	d^1	1	d^1	a^1	100

compatibility factors denoting affinity

**X^0 if X has not a misconception or
 X^1 is if X has a misconception**



Summary so far

- Bayes nets provide a natural representation for (causally induced) conditional independence
- Topology + CPTs = compact representation of joint distribution
- Generally easy for (non)experts to construct
- Canonical distributions (e.g., noisy-OR) = compact representations
- Continuous variables $\Rightarrow$ parameterized distributions (e.g., linear Gaussian)
- Bayes nets capture probabilistic influences, whereas causal networks capture causal relations and allow predicting effects of interventions
- Bayes nets instance of larger class of Probabilistic Graphical Models
- Various methods for learning probabilistic model structure and parameters

Suggested exercises from Russel & Norvig, 3rd Ed.

- 14.8 (car diagnosis), (a)-(d)
- 14.11 (nuclear plant)
- 14.12 (telescope problem)
- 14.14 (wrongful conviction)

Questions?